Harmonic zeros have a fourth-order scale cost

24 September 2026. This note strengthens the whole-filling criterion in Harmonic/contact Milnor matching. Its qualitative hypothesis that the actual harmonic representative have no zeros is replaced by a quantitative comparison between the original Einstein/calibration defect and a metric coordinate scale on the **entire filling**. No identification of a terminal ALE space with that filling is made.

The analytic ingredient is an elementary local Hodge estimate, composed with the inherited actual-form theorem. A classical local construction of Taubes supplies the complementary example: even a fixed actual monopole class, unchanged periods and boundary data, exact harmonicity, and vanishing averaged structure errors do not by themselves prevent zero circles. The example is not pointwise almost Einstein. These two statements specify exactly which additional scale information the present bridge uses.

1. A metric scale and a local theorem

Fix $0 < \alpha < 1$ and $Q \geqslant 2$. A number $0 < \rho \leqslant 1$ is an **admissible coordinate scale** at $x \in (M, g)$ if there is a smooth embedding $\phi : B_2(0) \subset \mathbb{R}^4 \longrightarrow M$, with $\phi(0) = x$, such that

$$\widehat{g} = \rho^{-2}\phi^* g, \qquad Q^{-1} I \leqslant \widehat{g} \leqslant QI, \qquad \|\widehat{g}\|_{C^{1,\alpha}(B_2)} \leqslant Q. \tag{1}$$

Changing the harmless numerical convention in the coordinate norm changes only the constants below. Write $U_{x,\rho} = \phi(B_2)$, and denote the supremum of these scales by $r_{\alpha,Q}(x)$. Smoothness supplies a positive scale at every point, after increasing the fixed Q if needed. The scale is capped at 1 because the target Einstein normalization has already fixed an ambient length unit. Harmonic coordinates satisfying (1) can be used, but their coordinate functions need not be harmonic.

Thus (1) is a condition solely on the metric, independent of the harmonic form or its zeros. It is not a lower bound on the injectivity radius of the entire manifold, and no uniform positive lower bound for this scale will be imposed on a sequence.

Theorem 1 (a harmonic zero costs normalized amplitude energy). There is $c = c(\alpha, Q) > 0$ with the following property. Let Ω be a smooth two-form satisfying

$$d\Omega = d_g^* \Omega = 0$$

on $U_{x,\rho}$, where (1) holds. If $\Omega(x) = 0$, then

$$\boxed{\int_{U_{x,\rho}} \left(|\Omega|_g - \sqrt{2}\right)^2 d\mu_g \geqslant c\rho^4.} \tag{2}$$

Self-duality is unnecessary for this local assertion. The normalization $\sqrt{2}$ is the one used by the global monopole construction.

Proof. Pull back and rescale both objects by

$$\widehat{g} = \rho^{-2}\phi^* g, \qquad \widehat{\Omega} = \rho^{-2}\phi^* \Omega.$$

Then $|\widehat{\Omega}|_{\widehat{g}} = |\Omega|_g \circ \phi$, the Hodge equations remain homogeneous, and the integral in (2) is ρ^4 times the corresponding integral on B_2. Work at this unit scale and call that integral A. If $A \geqslant 1$, the conclusion holds with any $c \leqslant 1$. Otherwise

$$\int_{B_2} |\widehat{\Omega}|^2 \, d\mu_{\widehat{g}} \leqslant 2A + 4\,\mathrm{Vol}_{\widehat{g}}(B_2) \leqslant C(Q). \tag{3}$$

Here is the regularity input, including its coefficient requirements. View $D = d + d^*_{\widehat{g}}$ on the full exterior-algebra bundle. Its principal symbol is Clifford multiplication and is uniformly invertible away from the zero covector. In coordinates its principal coefficients are uniformly $C^{1,\alpha}$, and its zero-order coefficients are bounded, by (1). The local first-order elliptic estimate on nested balls is

$$\|u\|_{W^{1,p}(B_a)} \leqslant C(p, Q, \alpha, a, b)\big(\|Du\|_{L^p(B_b)} + \|u\|_{L^p(B_b)}\big), \qquad 0 < a < b < 2. \tag{4}$$

One obtains (4) by freezing the principal coefficients on uniformly small coordinate balls, applying the constant-coefficient Fourier-multiplier estimate, absorbing their oscillation, and using a cutoff; the bounded zero-order term is retained on the right. A finite covering gives the displayed constants. This argument needs no bound on second metric derivatives and takes no derivative of Ricci curvature.

Apply (4) to the pure-degree form $\widehat{\Omega}$, viewed as an inhomogeneous form. Its full Hodge–Dirac image is zero. Starting from (3), successively use $W^{1,2} \hookrightarrow L^4$, the interior $W^{1,4}$ estimate, the critical embedding into each finite L^p, and the $W^{1,8}$ estimate. Morrey's inequality gives, on B_1,

$$\|\widehat{\Omega}\|_{C^{0,1/2}(B_1)} \leqslant C(\alpha, Q). \tag{5}$$

Since $\widehat{\Omega}(0) = 0$, uniform ellipticity and (5) imply $|\widehat{\Omega}|_{\widehat{g}} \leqslant 1/2$ throughout B_θ for a fixed $\theta = \theta(\alpha, Q) > 0$. Consequently

$$A \geqslant (\sqrt{2} - 1/2)^2 \operatorname{Vol}_{\widehat{g}}(B_\theta) \geqslant (\sqrt{2} - 1/2)^2 Q^{-2} |B_\theta| > 0.$$

Taking the minimum of this lower bound and 1 and rescaling proves (2). □

In particular, the theorem is not a new bounded-Ricci regularity theorem: it uses a declared coordinate scale. Its content for this project is that the actual Hodge equation turns the supplied mean amplitude into a quantitative obstruction to zeros at that scale.

2. Eliminating the auxiliary form from the condition

Use the input and conventions of Actual-form composition. Thus M is closed, connected, oriented and four-dimensional, and a is the first Chern class of a Spin^c structure whose unperturbed Seiberg–Witten equations are solvable for every metric in $[g]$. The structure remains fixed as the positive weight varies. Let

$$\begin{aligned} V &= \operatorname{Vol}(M, g), \qquad p = (a_g^+)^2 > 0, \\ r &= \fint_M |\operatorname{Ric}_g + g|^2, \\ q &= V^{-1}\left[\frac{3}{2}\int_M (s^2 + 12|W^+|^2) - 72\pi^2 p\right], \qquad \kappa = r + q \leqslant 1/2880. \end{aligned} \tag{6}$$

The curvature functional gives $q \geqslant 0$. Let H denote the harmonic self-dual representative of a_g^+, not the harmonic representative of the whole class when $a_g^- \neq 0$, and put $\Omega = -2\pi H$.

Lemma 2 (actual harmonic amplitude). The inherited supply theorem implies directly

$$\boxed{\int_M \big(|\Omega| - \sqrt{2}\big)^2 \, d\mu \leqslant 818\kappa V.} \tag{7}$$

There is no remaining auxiliary weight or negative-square budget in (7).

Proof. For every $\eta > 0$ sufficiently small, the same actual class supplies a smooth form ω satisfying

$$\int |\omega - \Omega|^2 \leqslant (49\kappa + \eta)V, \qquad \int (|\omega|^2 - 2)^2 \leqslant (720\kappa + \eta)V.$$

The reverse triangle inequality and

$$(t - \sqrt{2})^2 = \frac{(t^2-2)^2}{(t+\sqrt{2})^2} \leqslant \tfrac{1}{2}(t^2-2)^2, \qquad t \geqslant 0,$$

give

$$\int (|\Omega| - \sqrt{2})^2 \leqslant 2\int |\Omega - \omega|^2 + \int (|\omega|^2 - 2)^2 \leqslant (818\kappa + 3\eta)V.$$

The left side is independent of the weight and of η. Let $\eta \downarrow 0$. No convergence of the auxiliary forms is asserted or needed. □

Combining (2) and (7) yields a purely raw-input consequence:

$$\boxed{H(x) = 0 \quad \Longrightarrow \quad r_{\alpha,Q}(x)^4 \leqslant C(\alpha, Q)\kappa V.} \tag{8}$$

Indeed apply (2) at every admissible scale and then take their supremum. This also handles the case where that supremum is not attained.

3. Whole-core matching and exact realization

Retain the raw compactification hypotheses

$$\operatorname{diam}(M_j, g_j) \leqslant D, \qquad (a_j^+)^2 \geqslant A_0 > 0, \qquad \delta_j + q_j \longrightarrow 0, \quad \delta_j = \|\operatorname{Ric}^\sharp_{g_j} + I\|_{\infty,\mathrm{op}}. \tag{9}$$

Each a_j is supplied by an actual conformally solvable Spin^c structure, with the fixed-structure/variable-weight quantifier of Section 2. By Orbifold smoothing bridge, Theorem A, a subsequence has a compact negative KE orbifold limit (X, h, J). Its regular exterior, with fixed small collars around the singular links, embeds in M_j. Let $B_{\nu,j}$ be the complementary **whole** connected fillings, with their actual markings. As in that theorem, each chosen link bounds its indicated filling. All nested necks and bubbles are included in $B_{\nu,j}$.

For each j, suppose metric data supply a number $\rho_j > 0$ such that every point of $\bigcup_\nu B_{\nu,j}$ has an admissible coordinate scale at least ρ_j, with fixed α, Q. The coordinate neighborhoods lie in M_j and may cross the fixed boundary collars. This avoids any boundary estimate for the Hodge equation.

Theorem 3 (defect-to-core-scale realization). There is $c_0 = c_0(\alpha, Q) > 0$ such that the following conditions, for all sufficiently large indices in the extracted subsequence, force marked Milnor matching:

1. $(r_j + q_j)V_j < c_0\rho_j^4$;
2. each singular filling satisfies either
   - its limiting singularity is A_1 and $b_2(B_{\nu,j}; \mathbb{Q}) = 1$, or
   - $B_{\nu,j}$ is a rational homology four-ball.

The second alternative forces the limiting singularity to be Wahl, $\frac{1}{n^2}(1, na-1)$, with $n \geqslant 2$, $0 < a < n$, and $(a, n) = 1$. Every actual marked filling is diffeomorphic to the corresponding local Q-Gorenstein Milnor fiber.

If, in addition, the complex orbifold X admits a projective Q-Gorenstein smoothing, then each such M_j, on its original smooth structure, admits a negative Kähler–Einstein metric. A sufficient global smoothing test is $H^2(Z, T_Z(-\log E)) = 0$, with the resolution and divisor as in Lee–Park's criterion used in the inherited bridge.

The first condition is implied by either of the more directly recorded conditions

$$\frac{(r_j + q_j)V_j}{\rho_j^4} \longrightarrow 0, \qquad \text{or} \qquad \frac{(4\delta_j^2 + q_j)V_j}{\rho_j^4} \longrightarrow 0. \tag{10}$$

Proof. Since $r_j \leqslant 4\delta_j^2$, (9) puts the sequence within the range of Lemma 2. If H_j vanished at a point of a whole filling, (2) and (7), applied at the supplied scale, would imply $c\rho_j^4 \leqslant 818(r_j + q_j)V_j$. Choose $c_0 < c/818$. Thus every H_j is zero-free on the entire filling.

The inherited harmonic/contact theorem now applies, with no change to its remaining hypotheses. For clarity, its implications are as follows. The actual $\Omega_j = -2\pi H_j$ converges to the limiting Kähler form in $C^{1,\beta}$ on each fixed regular collar. Therefore it gives the canonical strong convex contact boundary. Self-duality and closedness, together with the newly proved nonvanishing throughout the core, make $(B_{v,j}, \Omega_j)$ a strong symplectic filling. Classical filling classification gives the A_1 Milnor filling in the first case, and a Wahl rational ball in the second. The boundary mapping-class extension argument in that theorem gives the actual relative marking, not merely an unmarked diffeomorphism.

If a global smoothing exists, its exterior trivialization and these relative diffeomorphisms glue to an oriented diffeomorphism from M_j to a smooth fiber. The canonical divisor is ample, by the negative KE metric on X and openness of ampleness in the Q-Gorenstein family. Aubin–Yau gives a negative KE metric on the smooth fiber; pull it back. This proves existence on M_j, without an Einstein inverse or a claim that the new metric is close to g_j. □

There is no positive lower bound on ρ_j: the theorem permits concentration at arbitrarily small scales if the original total defect decays faster than the fourth power of the smallest supplied core scale. The metric scale, the Betti/rational-ball condition, and the global smoothing test are separate verifiable inputs; none is declared to follow from (9) alone. The result includes non-Einstein inputs: sufficiently small smooth perturbations of any negative KE metric with its compatible actual class satisfy the criterion on prescribed cores, by continuity of the Hodge projection and the functionals. On a degenerating KE sequence, one may choose the perturbations separately at each positive ρ_j. This observation establishes nonemptiness, not a new construction of KE metrics or a new smoothing example.

Corollary 4 (non-Wahl rational cores have a quantitative cost). In (9), suppose a whole filling is a rational homology ball but its limiting singularity is not Wahl. Then for large j it contains an interior zero x_j of H_j, and

$$r_{\alpha,Q}(x_j)^4 \leqslant C(\alpha, Q)(r_j + q_j)V_j. \tag{11}$$

If all its points have a supplied scale at least ρ_j, then $(r_j + q_j)V_j \geqslant c_0\rho_j^4$, after reducing c_0.

Proof. The inherited contact-filling obstruction forces a zero, and its regular collar contains none. Apply (8). □

4. Unchanged periods and harmonicity do not remove the scale issue

The following construction is classical at its source. Its quantitative rescaling identifies a genuine limitation of replacing the new condition by averaged form information alone.

Proposition 5 (small-support harmonic zero circles). Let (S, g_0, J, Ω_0) be a fixed closed Kähler surface, with $|\Omega_0|_{g_0} = \sqrt{2}$. There are smooth metrics g_ρ and closed self-dual forms Ω_ρ, for $\rho \downarrow 0$, such that:

1. $g_\rho = g_0$ and $\Omega_\rho = \Omega_0$ outside a fixed Darboux ball of radius $C\rho$;
2. $[\Omega_\rho] = [\Omega_0]$, and Ω_ρ is its **actual harmonic representative** for g_ρ;
3. Ω_ρ has a transverse zero circle in that ball;
4. g_ρ is uniformly bilipschitz to g_0, $\|\Omega_\rho\|_\infty \leqslant 2\sqrt{3}$, and

$$\begin{aligned} \int_S (|\Omega_\rho|^2 - 2)^2 \, d\mu_{g_\rho} &= O(\rho^4), \\ \int_S |\Omega_\rho - \Omega_0|^2_{g_\rho} \, d\mu_{g_\rho} &= O(\rho^4), \\ \int_S |\nabla^{g_\rho} \Omega_\rho|^2 \, d\mu_{g_\rho} &= O(\rho^2), \\ \int_S |\nabla^{2,g_\rho} \Omega_\rho|^2 \, d\mu_{g_\rho} &= O(1). \end{aligned} \tag{12}$$

In particular the first three integrals tend to zero, although zeros persist. The volumes stay bounded above and below, so the same conclusions hold for averaged integrals. The Hodge Laplacian of each form is exactly zero.

Proof. Taubes [T06], Proposition 2.1 and its proof in Section 2, construct a near-symplectic form on a standard symplectic ball which equals the standard form outside a compact interior set and has one transverse zero circle. Its compactly supported difference from the standard form is exact: $H^2_c(B^4; \mathbb{R}) = 0$. Fix one such model, denoted $\widetilde{\Omega}$, on a ball of fixed radius.

Auroux–Donaldson–Katzarkov [ADK05], Proposition 1, supply a smooth metric making this closed form self-dual and hence harmonic. The metric may be prescribed to be Euclidean near the boundary, where the form is standard. To see the relative assertion directly from their Section 7.2, positive three-plane bundles containing a given positive two-form form a convex set in the graph parametrization of the positive Grassmannian. Extend the positive plane determined near the zero circle, and splice it to the Euclidean positive plane on the exterior annulus, where the form is nonzero. This constructs a conformal class agreeing with the prescribed one there. Choose a smooth conformal factor agreeing with 1 there and large enough on the interior to arrange $|\widetilde{\Omega}| \leqslant 2\sqrt{3}$. This rescaling preserves self-duality of two-forms in dimension four. We may also arrange that its Ricci tensor is nonzero at some interior point: a further smooth conformal perturbation supported away from the zero circle can prescribe a nonzero Hessian at a point, and the conformal Ricci formula then ensures this. Take the perturbation small enough to retain the amplitude bound, first leaving a strict margin there. Fix the resulting smooth model metric.

Take Darboux coordinates for Ω_0 at a point of S, with g_0 equal to the Euclidean metric at the origin. Transplant the model by dilation $x = \rho y$: its coordinate coefficients are $\tilde{g}(x/\rho)$ and $\widetilde{\Omega}_{ab}(x/\rho)$. Equivalently, the pullbacks under dilation are $\rho^2 \tilde{g}$ and $\rho^2 \widetilde{\Omega}$. The form equals Ω_0 near the boundary and glues exactly. On a surrounding annulus keep that form fixed and interpolate the compatible positive planes from the model metric to those of g_0. Since $g_0(\rho y)$ is uniformly smooth and tends to its value at the origin, this relative interpolation has uniformly bounded derivatives in the y

coordinates. Its conformal factor can again enforce the stated amplitude bound and match g_0. Both objects are unchanged outside the enlarged ball.

The compactly supported exactness proves equality of classes. Closedness and self-duality give $d^*_{g_\rho}\Omega_\rho = 0$; on the closed manifold Hodge uniqueness makes it the actual harmonic representative. The inserted transverse zero circle is unaffected by the outer annulus.

In rescaled coordinates the metrics are uniformly smooth and elliptic, the form is uniformly smooth, and the changed region has volume $O(\rho^4)$. One and two covariant derivatives of the form are respectively $O(\rho^{-1})$ and $O(\rho^{-2})$ there. Outside, the original Kähler form is parallel for the original metric. These facts prove all four estimates (12). □

This applies with an actual class, rather than merely a real cohomology class. For example take the negative KE product of two closed hyperbolic curves of genus at least two. Put $a = c_1(S,J)$, so $[\Omega_0] = -2\pi a$, and $a^2 > 0$. Its canonical Spin^c class and its negative are ordinary monopole classes, by the classical Seiberg–Witten input recalled in LeBrun [L06], Section 3. In particular the same structure is solvable for all metrics, hence for all conformal weights for each g_ρ. Proposition 5 gives

$$a^+_{g_\rho} = a, \qquad a^-_{g_\rho} = 0, \qquad -2\pi H_{g_\rho} = \Omega_\rho,$$

with $p = a^2$ fixed and positive. All periods, the positive projection budget, and the negative-square cost are unchanged; there are nevertheless harmonic zero circles and arbitrarily small averaged structure errors.

The construction is also relative on any prescribed symplectic Milnor filling: perform it in an interior Darboux ball of an A_1 or Wahl fiber, away from its contact collar. The filling, its boundary marking, its boundary form and **every period on every closed two-cycle** are unchanged. It creates interior zeros of a closed self-dual harmonic form. On a filling this means a solution of the interior Hodge equations; no uniqueness for an unspecified boundary value problem is claimed. On a closed ambient manifold the uniqueness assertion is the one in Proposition 5. No zero-free hypothesis can be deduced from these period and collar data alone.

At a zero of the inserted model, (1) holds with scale at least $c_1\rho$, for fixed α, Q. Theorem 1 supplies the reverse order of magnitude for the amplitude error, while (12) supplies the upper bound. Thus the power four in (2) is the correct scaling for the local statement: a hypothesis $A = o(\rho^\gamma)$ with $\gamma < 4$ would allow these zero-circle examples. This is not a claim of sharp constants or sharpness within pointwise almost-Einstein sequences.

5. Exact scope of the new bridge

The conclusion added to the previous theorem is

actual raw defect below the fourth power of the whole-core scale $\implies$ marked Milnor matching

in its stated A_1/rational-ball topological regimes. After the separate global Q-Gorenstein test this gives an Einstein metric on the original smooth manifold, as required by Yau26 for the covered inputs.

The zero-circle construction explains why keeping only global averages would lose relevant information. It does **not** satisfy the claimed small pointwise Ricci hypothesis (9): the fixed nonzero model Ricci tensor in the preceding proof gives $\delta_-(g_\rho) \geqslant c\rho^{-2} - 1$ at its transplanted point. Thus the pointwise Ricci defect actually diverges. Nor do (12) establish that the raw functional $r_\rho + q_\rho$ tends

to zero. Therefore it neither disproves automatic matching under the complete original input nor supplies a Yau26 counterexample. What it disproves is inference of nonvanishing from the listed harmonic, period and average-form properties alone.

The remaining mathematical question is consequently specific: does the pointwise Ricci equation plus actual near-saturation force the ratio in (10), or provide a different zero-removal mechanism on the concentration cores? That implication has not been proved here. The present theorem allows the ratio to be checked directly and quantifies its failure when contact topology forces a zero; it does not assume that all relevant ALE information has already classified the whole core.

References and inherited dependencies

- [T06] C. H. Taubes, *A proof of a theorem of Luttinger and Simpson about the number of vanishing circles of a near-symplectic form on a 4-dimensional manifold*, Mathematical Research Letters **13** (2006), no. 4, 557–570. Proposition 2.1 and the compactly supported birth construction in Section 2. Publisher PDF.
- [ADK05] D. Auroux, S. K. Donaldson and L. Katzarkov, *Singular Lefschetz pencils*, Geometry & Topology **9** (2005), 1043–1114. Proposition 1 and its positive-plane-bundle proof in Section 7.2. Original manuscript.
- [L06] C. LeBrun, *Four-Manifolds, Curvature Bounds, and Convex Geometry*, preprint (2006). Section 3 for actual monopole classes, and Section 2 for Hodge and chirality conventions. Original manuscript.
- [Supply] Actual-form composition, the inherited theorem, estimates (2) and (4). Their constants give 818 in (7); their proof is not replaced by this note.
- [Compactification] Orbifold smoothing bridge, Theorems A and C, and Corollary D. These supply the bounded-Ricci compactification and the separately checked global smoothing input.
- [Matching] Harmonic/contact Milnor matching, Theorem 1 and Corollaries 6–7, including its classical filling classifications and boundary-extension proofs. Only its whole-core nonvanishing premise is replaced here.

No priority claim is made for the local elliptic estimate or the zero-circle construction. The new project-level application is their composition with actual monopole supply and marked filling classification.